# LLM6G: A Shift-Aware Forecast-to-Control Pipeline for 6G Mobile Network Traffic

Colin Minini, Tariq Barki, Alexios Aravanis
CentraleSupelec, Paris-Saclay University

***Abstract*—Time series forecasting is a key enabler for proactive network management, supporting resource provisioning, energy-aware operation, and service-level compliance. In mobile networks evolving toward 6G, traffic demand progresses through regimes separated by distribution shifts — daily cycles, event surges, and broader demand trends — so that a controller reacting only to current load risks unsafe or wasteful operating points when the next regime arrives.**

**Traditional statistical models provide interpretable baselines but assume stationarity. Deep learning approaches have advanced forecast accuracy on non-stationary telecom data, yet existing work typically optimizes forecasting in isolation: network control papers set provisioning policies around predicted load levels without explicitly modeling when the next distribution break will occur. No end-to-end pipeline exists that makes the regime boundary a first-class control object evaluated under shared metrics on public data.**

**We propose LLM6G, a shift-aware forecast-to-control pipeline where any probabilistic forecaster plugs into a shared interface: detect the next distribution break on the median forecast, compute a safe ceiling from the upper quantile up to that break, and refresh at the predicted break. The pipeline integrates Chronos-2, an LLM-based zero-shot time-series foundation model, into a 6G-oriented control loop. Five forecasters are benchmarked on the public TIM Milan dataset at its native 10-minute cadence.**

**Coverage and sharpness compete: no model dominates both. TFT achieves the best trained operating point, enabled by channel-selective instance normalization. Chronos-2, with no domain training, delivers the tightest safe ceiling across all models, demonstrating that LLM-based foundation models can operate directly in 6G control loops without retraining. Code and benchmark protocol are publicly available.**

***Index Terms*—traffic forecasting, change-point detection, probabilistic forecasting, network control, TIM Milan, Chronos-2, Temporal Fusion Transformer, 6G**

## I. Introduction

Time series forecasting plays a fundamental role in a wide array of real-world applications, from financial market prediction and energy demand estimation to healthcare monitoring and environmental modeling [1]. Its importance has grown significantly in the domain of Software-Defined Networking and Network Function Virtualization, where forecasting is a critical enabler for intelligent controllers and autonomous service orchestration. Accurate predictions of future demand allow the network to allocate resources proactively rather than reactively, improving both efficiency and quality of service [2].

This capability is especially critical in mobile networks evolving toward 6G, where traffic demand exhibits complex and heterogeneous patterns. Proactive resource allocation, dynamic network slicing, and energy-aware base-station management all depend on anticipating how traffic will evolve over the coming hours [3]–[5]. In practice, traffic demand progresses through regimes — daily cycles, weekends, event surges, and broader seasonal trends — that shift both the level and the shape of the traffic distribution. These regime changes matter directly for control: a provisioning policy built around the current load level can leave the network at an unsafe or wasteful operating point when the next regime arrives.

Classical statistical models such as ARIMA [6] and exponential smoothing [1] have historically provided interpretable and mathematically tractable solutions for time series forecasting. However, these methods typically assume linear relationships and stationarity — constraints that limit their ability to capture the complex, non-linear, and dynamic behaviors observed in modern network environments. When traffic undergoes abrupt regime changes, stationary models can produce misleading forecasts precisely when accuracy matters most.

To address these limitations, deep learning approaches have advanced the state of the art by modeling long-range temporal dependencies and non-linear dynamics. Recurrent architectures such as LSTMs, attention-based models like the Temporal Fusion Transformer (TFT) [7], and autoregressive probabilistic models like DeepAR [8] have demonstrated strong results on telecom traffic prediction tasks [9]. More recently, the foundation-model paradigm has reached time series: Chronos [10] and Chronos-2 [11] are pretrained on large diverse corpora and can be deployed without target-specific retraining, offering an operationally attractive alternative in networks where recurring traffic shifts make periodic retraining costly.

Despite these advances, existing work addresses the forecasting-to-control problem in parts. Traffic forecasting papers optimize accuracy metrics — RMSE, pinball loss [12], coverage [13] — but stop before the control decision [2], [3]. Network control papers set sleeping or provisioning policies around predicted load levels without explicitly modeling when the next distribution break will occur [14]–[16]. What is missing is an end-to-end shift-aware pipeline that makes the next regime boundary a first-class control object, evaluated on public data under shared metrics. The relevant control quantity is not the next-step forecast; it is the next stable operating interval and the safe ceiling over that interval.

This paper fills that gap. We present LLM6G: a general forecast-to-control pipeline where any probabilistic forecaster plugs into a shared interface. The pipeline forecasts a probabilistic traffic path, detects the most prominent distribution break on the median forecast, computes a safe ceiling from the upper quantile up to that break, and refreshes after the predicted break. The name reflects the two central components: the LLM (Chronos-2, a foundation model for time series) and the 6G control objective. We use "LLM" in the broad sense of large pre-trained generative models for sequential data, consistent with the emerging literature on LLMs for time series [10], [11]. By explicitly separating forecast quality from control quality, the pipeline enables a principled comparison of forecasters on the metric that matters for provisioning: how tight and how safe is the resulting ceiling.

We benchmark LLM6G on the public TIM Milan dataset [17] at its native 10-minute cadence. The dataset remains active in the public forecasting literature [18] and provides a clean non-augmented telecom traffic proxy at city scale. Five forecasters are evaluated inside the pipeline: TFT, LSTM, DeepAR, Chronos-2, and Seasonal Naive. Coverage and sharpness [19] of the safe ceiling are the primary control metrics. Neither dominates: a tighter ceiling reduces over-provisioning but risks coverage violations. TFT achieves the best trained operating point. Chronos-2 achieves the tightest ceiling across all five models with no domain training, supporting its role as a zero-shot foundation model for 6G control under recurring distribution shifts.

The paper makes four contributions:

- **LLM6G pipeline.** A general model-agnostic forecast-to-control framework: probabilistic forecast, break-point detection, pre-break safe ceiling, receding-horizon refresh. Any forecaster that produces $q_{0.5}$ and $q_{0.95}$ plugs in.
- **TFT as best trained model.** Channel-selective instance normalization enables TFT to generalize across distribution shifts. Without it, the validation-to-test RMSE gap exceeds 30%.
- **Chronos-2 in the control loop.** First evaluation of Chronos-2 in a shift-aware forecast-to-control pipeline. It achieves the tightest safe ceiling across all five models with no domain training.
- **Public TIM Milan benchmark.** 200-cell trainable subset, 144-step daily-cycle context, native 10-minute cadence, 5-model evaluation under a shared canonical protocol.

The remainder of this paper is organized as follows. Section II reviews background and related work. Section III describes the LLM6G pipeline and formalizes the evaluation metrics. Section IV details the forecasters and implementation decisions. Section V presents the experimental setup and dataset. Section VI reports and discusses results. Section VII concludes and outlines future work.

## II. Background and Related Works

This section reviews the three lines of work that converge in the LLM6G pipeline: telecom traffic prediction on public benchmarks, forecasting-driven network control, and neural probabilistic forecasting including foundation models and change-point detection. For each, we identify the contributions and the gap that remains.

### A. Telecom traffic prediction and public benchmarks

Public telecom traffic benchmarks remain limited. The TIM Milan release by Barlacchi *et al.* [17] is one of the most reusable public datasets for city-scale telecom activity. Recent work continues to use it to evaluate stacked recurrent models, transformers, and TimeGPT-style forecasters [18]. Surveys of cellular traffic prediction confirm the same point: the forecasting literature is broad, but reproducible public benchmarking is thinner than proprietary operator evaluation [2], [3]. Several papers apply spatial-temporal deep models on grid-level data [9], [20], [21]. Our paper keeps the public-benchmark perspective and shifts the evaluation criterion from forecasting accuracy alone to a control loop centered on the next predicted regime change.

While these works advance forecast accuracy, most stop before connecting predictions to a control decision. The next subsection reviews work that takes that step.

### B. Forecasting for network control and energy-aware operation

Traffic prediction is a standard ingredient in energy-aware network control. Recent work has applied it to green cellular networking [4], [5], [22], base-station sleeping [14], [15], and digital-twin-assisted management [16]. These papers motivate the operational value of forecasting. They typically optimize around predicted load levels or sleeping decisions, however, rather than around the next predicted distribution break. Our pipeline sits one level above point prediction: it explicitly predicts the next regime boundary and sets a safe ceiling only over the resulting stationary interval.

The forecasting models used in these control settings range from classical to deep learning. The next subsection reviews the specific architectures and paradigms relevant to the LLM6G pipeline.

### C. Neural probabilistic forecasters, foundation models, and change detection

DeepAR [8] models per-series distributions autoregressively and extracts quantiles from Monte Carlo samples at inference. The Temporal Fusion Transformer [7] adds multi-head self-attention with variable selection networks and interpretable gating, achieving strong multi-horizon quantile forecasting; it has also been applied directly to network traffic prediction [9]. Both are trained baselines in this paper. Chronos [10] and Chronos-2 [11] bring the foundation-model paradigm to time series: pretrain on a large diverse corpus and deploy without target-specific retraining. Concurrent work applies time-series foundation models to ISP and optical network telemetry [3]; these focus on forecasting accuracy rather than control-oriented evaluation. For our setting, zero-shot deployment is operationally attractive because recurring traffic shifts can make periodic retraining costly.

Change-point detection provides a direct way to reason about regime boundaries. PELT [23] is an efficient exact change-point detector. CLASP [24] classifies sliding windows via k-nearest-neighbour accuracy, returns an explicit no-break decision when no score threshold is exceeded, and requires no assumption about the cost structure of change points. We select CLASP through a validation sweep and use it as the shared detector for all forecasters.

What is missing in the public telecom setting is an end-to-end benchmark connecting probabilistic forecasting, break detection, safe ceiling computation, and control-oriented evaluation under established metrics such as quantile loss [12], prediction interval coverage [13], and proper scoring rules for sharpness [19]. The contribution of this paper is that pipeline and its instantiation on a public benchmark. We do not propose a new forecasting architecture.

## III. The LLM6G Forecast-to-Control Pipeline

### A. Design philosophy

The pipeline is intentionally simple. Its value is not in the forecasting model — any calibrated forecaster plugs in. Its value is in making the next predicted distribution break a first-class control object. Existing control policies use forecasts to set a resource budget for the full horizon. This policy sets a tighter budget valid only over the next stable sub-interval and refreshes when that interval ends.

Five forecasters are benchmarked inside the same pipeline: TFT, LSTM, DeepAR, Chronos-2, and Seasonal Naive. They differ in architecture, training regime, and how they produce uncertainty estimates. All share the same output interface: a median path $q_{0.5}(1:H)$ and an upper path $q_{0.95}(1:H)$ over horizon $H$.

### B. Pipeline stages

Figure 1 shows the full pipeline. Given a traffic context $x_{t-C+1:t}$ of length $C$:

**Stage 1 — Probabilistic forecast.** The forecaster produces two paths over the next $H$ steps:

$$q_{0.5}(1:H) \quad \text{and} \quad q_{0.95}(1:H).$$

**Stage 2 — Break-point detection.** A CLASP change-point detector [24] is applied to the median path $q_{0.5}$. It returns the most prominent predicted break point $\tau_{\text{pred}}$ — the position that globally maximises the classification score separating the two forecast regimes — or $H$ if no significant break is detected.

**Stage 3 — Safe ceiling.** The pre-break safe ceiling is

$$c_{\text{safe}} = \max_{1\leq k\leq\tau_{\text{pred}}} q_{0.95}(k).$$

This ceiling covers the predicted stationary interval using the upper quantile path up to the predicted break.

**Stage 4 — Control and refresh.** The network is provisioned up to $c_{\text{safe}}$ until $\tau_{\text{pred}}$. At that point, the pipeline refreshes: it collects new observations, updates the context, and runs stages 1–3 again. This yields a receding-horizon loop driven by predicted regime changes.

### C. Why explicit break detection matters

Provisioning the network up to a ceiling computed from the full horizon overestimates the requirement over the early stationary part of that horizon. The ceiling should reflect only the traffic regime that is expected to be in place. By restricting the ceiling computation to $[1, \tau_{\text{pred}}]$, the pipeline produces a tighter and more accurate target as long as the break is predicted well.

This separates two objectives that are often conflated: forecast quality over the full horizon and control quality over the next stable interval. A model with good RMSE can still produce a poor control interval if its break point or upper quantile is misplaced.

### D. Evaluation metrics

Let $x_{t-C+1:t}$ denote a context window of length $C$ and let $H$ denote the forecast horizon. A forecaster receives the context and produces two paths over the next $H$ steps: the median path $q_{0.5}(1:H)$ and the upper path $q_{0.95}(1:H)$.

The control loop focuses on the most prominent predicted regime change within the horizon. Let

$$\tau_{\text{pred}} = \text{CP}(q_{0.5}(1:H))\,, \tag{1}$$

where $\text{CP}(\cdot)$ returns the globally highest-scoring change point on the median path, or $H$ if no break is detected. The pre-break safe ceiling is

$$c_{\text{safe}} = \max_{1\leq k\leq\tau_{\text{pred}}} q_{0.95}(k). \tag{2}$$

For evaluation, the same detector is applied to the realized future path to obtain the reference break $\tau_{\text{ref}}$. Coverage and sharpness are evaluated against $\tau_{\text{ref}}$: the safe ceiling is computed from the predicted break $\tau_{\text{pred}}$, but tested against the realized pre-break interval $\mathcal{I}_{\text{ref}} = x_{1:\tau_{\text{ref}}}$. This asymmetry is intentional — it measures whether the committed ceiling protects the network throughout the actual stable regime, regardless of when the pipeline predicted the break. The pipeline is assessed with four quantities:

- *Change-point timing error*: $\text{MAE}_{\text{CP}} = |\tau_{\text{pred}} - \tau_{\text{ref}}|$, averaged over windows.
- *Tolerance hit rate*: fraction of windows where $|\tau_{\text{pred}} - \tau_{\text{ref}}| \leq 3$ steps (30 minutes at 10-minute cadence).
- *Coverage rate* [13]: fraction of realized pre-change points that stay below $c_{\text{safe}}$.
- *Relative sharpness* [19]: average of $(c_{\text{safe}} - \max(\mathcal{I}_{\text{ref}}))/\max(\mathcal{I}_{\text{ref}})$ over windows. Expresses excess ceiling as a fraction of realized demand.

Formally:

$$\text{CoverageRate} = \frac{\sum_{x\in\mathcal{I}_{\text{ref}}} \mathbf{1}[x \leq c_{\text{safe}}]}{|\mathcal{I}_{\text{ref}}|}, \tag{3}$$

$$\text{RelSharpness} = \frac{c_{\text{safe}} - \max(\mathcal{I}_{\text{ref}})}{\max(\mathcal{I}_{\text{ref}})}. \tag{4}$$

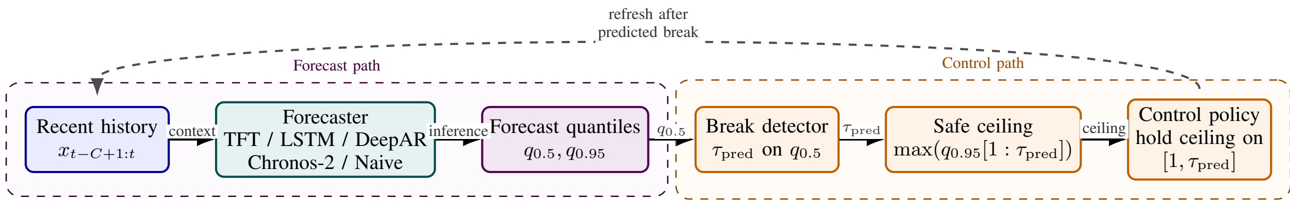


Fig. 1. The LLM6G forecast-to-control pipeline. Any forecaster that produces a median path $q_{0.5}$ and an upper path $q_{0.95}$ over horizon $H$ plugs into the shared interface. The five forecasters evaluated in this paper — TFT, LSTM, DeepAR, Chronos-2, Seasonal Naive — are interchangeable blocks at the left. The downstream controller and evaluation logic are identical across all models.

Coverage and relative sharpness are competing objectives. A forecaster that always sets a wide ceiling achieves high coverage at the cost of over-provisioning. A forecaster that sets a tight ceiling reduces wasted capacity but risks ceiling violations. Small positive relative sharpness is ideal. Negative relative sharpness is a control failure: the ceiling already lies below the realized traffic peak before the break.

We report both metrics without collapsing them. The trade-off is a property of the forecaster.

## IV. Forecasters and Implementation

All forecasters share the same output interface: a median path and an upper path over horizon $H = 48$ steps. This section describes each model and documents the implementation decisions that had measurable impact on control performance.

### A. Temporal Fusion Transformer

TFT [7] is the best-performing trained model in this benchmark. Its architecture combines variable selection networks (VSN) that learn which inputs are relevant at each step, an encoder-decoder LSTM (hidden size 128, 2 layers, dropout 0.1), interpretable multi-head self-attention (4 heads, shared value projection, causal mask), gated residual networks (GRN) for feature enrichment, and a direct quantile output head producing $q_{0.5}$ and $q_{0.95}$.

Time features — cyclic encodings of hour-of-day and day-of-week for the 10-minute cadence — are concatenated with the raw traffic value to form a multi-channel input tensor.

**Channel-selective instance normalization.** TFT receives a multi-channel input (raw value on channel 0; time features on channels 1+). Time features encode calendar position and must not be rescaled. We normalize only the raw-value channel:

$$\text{scale} = \text{mean}(|x_{1:C,0}|) + 1, \tag{5}$$

divide channel 0 by *scale* before encoding, and rescale the output by *scale* before computing the loss. The $+1$ prevents division by zero on constant or sparse series.

Without this fix, TFT memorizes absolute traffic levels instead of learning scale-invariant dynamics. The validation-to-test RMSE gap exceeds 30% without normalization, because the test period has different average traffic magnitudes than the validation period. With normalization, the gap closes to under 2%.

Training uses pinball loss at $q \in \{0.5, 0.95\}$. The checkpoint monitor is $\text{pinball}_{0.5} + \text{pinball}_{0.95}$ on the validation set.

### B. Chronos-2 (LLM baseline, zero-shot)

Chronos-2 [11] is a foundation model for time-series forecasting based on a patch-based encoder-only transformer with a direct quantile head. It is pretrained on a large diverse corpus of public and synthetic time series and deployed without any Milan-specific fine-tuning. Quantile forecasts are produced directly by the quantile head in a single forward pass over 21 quantile levels.

In the LLM6G framing, Chronos-2 is the LLM component: a broadly trained model deployed directly into a 6G-oriented control loop. The key operational property is that it requires no retraining when the traffic distribution shifts — a practical advantage in networks that undergo recurring regime changes. We use the `amazon/chronos-2` checkpoint (120M parameters).

### C. LSTM (trained baseline)

The LSTM forecaster uses a 2-layer LSTM (hidden size 128, dropout 0.1) followed by a direct quantile output head that produces all $H \times 2$ outputs in a single forward pass. Instance normalization is applied to the context:

$$\text{scale} = \text{mean}(|x_{1:C}|) + 1. \tag{6}$$

The context is divided by *scale* before encoding; outputs are rescaled accordingly. Training uses pinball loss.

### D. DeepAR (cautionary baseline)

DeepAR [8] uses an autoregressive LSTM with per-series item embeddings (dimension 20), time features (hour-of-day, day-of-week, log-age), and a Gaussian parameter head. Quantile forecasts are extracted from 200 Monte Carlo samples at inference.

**Likelihood choice.** DeepAR supports multiple likelihoods, including Gaussian for real-valued data and Negative Binomial for count data. Milan traffic values are continuous grid-cell aggregates; we use Gaussian likelihood as appropriate.

The consequence, however, is that Gaussian confidence intervals prove too narrow for the heavy-tailed traffic distribution: q95 coverage on the test set is 0.667, and relative sharpness is $-0.041$ (the ceiling falls 4.1% below the realized demand peak). A negative relative sharpness means the safe ceiling falls below the realized traffic peak before the break — the pipeline under-provisions rather than over-provisions. This is a control failure. We include DeepAR as a documented case of correct architecture with a likelihood mismatch for this data type.

**Checkpoint monitoring.** Initial experiments used Gaussian NLL as the early-stopping monitor. NLL triggered early stopping before coverage had matured. Switching the monitor to $\mathrm{pinball}_{0.5} + \mathrm{pinball}_{0.95}$ extends training to a later checkpoint where the quantile paths have properly converged, recovering substantially better coverage.

### *E. Seasonal Naive (lower baseline)*

The Seasonal Naive forecaster requires no training. The median forecast is the lag-144 value (yesterday-at-the-same-time). The upper-path forecast is the median plus the 95th percentile of historical differences $\{x_t - x_{t-144}\}$ computed over the context window. This provides a simple operational heuristic and an interpretable lower bound.

### *F. CLASP change-point detector*

CLASP [24] detects the most prominent break point non-parametrically. For each candidate $\tau$, it classifies sliding windows as pre-change or post-change using k-nearest-neighbour accuracy on z-normalized window features. It returns the $\tau$ with the maximum classification score if that score exceeds a threshold; otherwise it returns "no break." This explicit no-break decision is important: a forecaster that never predicts a break applies the ceiling to the full horizon, which is correct when the traffic regime is stable.

CLASP is selected over PELT because it produces an explicit no-break output via score thresholding and requires no assumption about the cost structure of change points. Parameters are tuned via a validation sweep over 36 combinations (min_size $\in \{8, 10, 12\}$, period_length $\in \{8, 16, \mathrm{auto}\}$, score_threshold $\in \{0.5, 0.6, 0.7, 0.75\}$). The best configuration is shared across all models: min_size $= 8$, period_length $= 16$, score_threshold $= 0.5$. Note that break F1 saturates at 1.0 on training data for all configurations; selection is therefore based on $\tau$-MAE rather than F1.

## V. Experimental Evaluation

### *A. TIM Milan benchmark*

The benchmark is built from the TIM Milan telecom activity dataset released by Barlacchi *et al.* [17]. Raw archives are accessed through the public O-RAN SC dataset mirror. Each record contains a Milan grid-cell identifier, a timestamp, and an `InternetTraffic` value. We use only this field.

The benchmark keeps the original 10-minute cadence. The harness concatenates daily archives, aligns the city-wide time grid, and retains only fully observed cells. No imputation is applied. The complete-cell benchmark contains 8,883 cells and 8,928 timestamps from 2013-11-01 through 2014-01-01.

For the experiments in this paper, a trainable subset of 200 cells is used. The subset is neutral: it keeps the first 200 complete cells by ascending grid-cell identifier. This rule avoids activity-based cherry-picking.

**Context length.** The context is set to $C = 144$ steps, which spans exactly one full daily cycle at 10-minute cadence. A 24-hour context window provides the prior day's complete traffic pattern as input. Shorter contexts miss the dominant periodicity; longer contexts risk including multiple regime changes in the conditioning window.

TABLE I
TIM Milan benchmark configuration.

| Quantity | Value |
|---|---|
| Complete grid cells | 8,883 |
| Trainable subset | 200 cells |
| Total timestamps | 8,928 |
| Cadence | 10 min |
| Date range | 2013-11-01 to 2014-01-01 |
| Context length $C$ | 144 steps (24 h) |
| Horizon $H$ | 48 steps (8 h) |
| Train / val / test rows | 6,249 / 893 / 1,786 |
| Test windows | 7,400 |

TABLE II
Model roles and training regimes.

| Model | Training | Role |
|---|---|---|
| TFT | supervised | best trained |
| LSTM | supervised | trained baseline |
| DeepAR | supervised | cautionary |
| Chronos-2 | zero-shot | LLM baseline |
| Seasonal Naive | none | lower bound |

The dataset is split chronologically in 70/10/20 proportion, yielding 6,249 train rows, 893 validation rows, and 1,786 test rows. All models share the same split. Evaluation uses non-overlapping horizon-step windows (step $= 48$) anchored at split start, yielding 7,400 test windows (200 series $\times$ 37 windows) and 3,600 validation windows. All models are evaluated on identical windows, removing sampling variance when comparing trained and zero-shot forecasters.

The benchmark represents public urban telecom activity and not operator base-station counters. All claims are at the level of grid-cell telecom proxies.

### *B. Trained model hyperparameters*

TFT, LSTM, and DeepAR share the recurrent backbone: two layers, hidden size 128, dropout 0.1. TFT adds 4-head interpretable self-attention with a shared value projection and a causal mask. All three models output quantiles at $\{0.5, 0.95\}$.

LSTM and TFT are trained with the Adam optimizer and pinball loss [12] at $q \in \{0.5, 0.95\}$. DeepAR uses Gaussian NLL as its training loss, consistent with its probabilistic output head. The early-stopping monitor for all three models is

$$\mathcal{L}_{\mathrm{monitor}} = \mathrm{pinball}_{0.5} + \mathrm{pinball}_{0.95}$$

evaluated on the validation set. Training runs for a maximum of 30 epochs with early stopping patience of 3 epochs, evaluated every epoch. The checkpoint with the lowest monitor value is retained.

### *C. Chronos-2 setup*

Chronos-2 is evaluated in zero-shot mode using the `amazon/chronos-2` checkpoint (120M parameters) with

TABLE III
FORECAST METRICS ON THE TIM MILAN TEST SET (7,400 WINDOWS). LOWER IS BETTER FOR RMSE AND PINBALL. COVERAGE TARGET: 0.95.

| Model | $\mathrm{RMSE}_{q50}$ | $\mathrm{Pinball}_{q50}$ | $\mathrm{Pinball}_{q95}$ | $\mathrm{Cover}_{q95}$ |
|---|---|---|---|---|
| TFT | **5.658** | **1.152** | **0.357** | 0.924 |
| LSTM | 5.684 | 1.389 | 0.385 | **0.951** |
| DeepAR | 5.796 | 1.493 | 0.817 | 0.667 |
| Seasonal Naive | 7.263 | 1.405 | 0.489 | 0.858 |
| Chronos-2 | 7.071 | 2.108 | 0.728 | 0.814 |

no fine-tuning on Milan data. Quantile forecasts are produced directly by the model's quantile head. No adaptation is applied.

### D. CLASP detector configuration

CLASP parameters are tuned on validation data via a grid search over 36 combinations:

- `min_size` $\in \{8, 10, 12\}$
- `period_length` $\in \{8, 16, \text{auto}\}$
- `score_threshold` $\in \{0.5, 0.6, 0.7, 0.75\}$

The selection criterion is $\tau$-MAE on validation windows where both the predicted and true break fall within the horizon. Break F1 saturates at 1.0 on training data for all configurations and provides no discrimination. The best configuration — min_size $= 8$, period_length $= 16$, score_threshold $= 0.5$ — is shared across all five forecasters.

### E. Metrics

Forecast quality is measured by RMSE [1] on the median path, pinball loss [12] at $q \in \{0.5, 0.95\}$, and q95 coverage [13] (fraction of realized values below $q_{0.95}$).

System quality is measured by change-point timing MAE ($\mathrm{MAE}_{\mathrm{CP}}$), tolerance hit rate (fraction of windows with $|\tau_{\mathrm{pred}} - \tau_{\mathrm{ref}}| \leq 3$ steps), pre-break coverage rate (fraction of realized pre-break values below $c_{\mathrm{safe}}$), and relative sharpness ($c_{\mathrm{safe}} - \max(\mathcal{I}_{\mathrm{ref}})$). The tolerance threshold of 3 steps corresponds to 30 minutes at the native 10-minute cadence. These metrics follow established practices for probabilistic forecast evaluation [19].

## VI. RESULTS AND DISCUSSION

### A. Forecast quality

Table III and Figure 2 show forecast results across all five models. TFT leads on point accuracy (RMSE 5.658, $\mathrm{pinball}_{q50}$ 1.152) and upper-path precision ($\mathrm{pinball}_{q95}$ 0.357). LSTM is close on RMSE (5.684) and achieves the strongest coverage (0.951, meeting the 0.95 target). Seasonal Naive is worst on RMSE (7.263) but reaches reasonable coverage (0.858).

The two failure modes appear in the remaining models. DeepAR achieves competitive RMSE (5.796) but its Gaussian intervals are too narrow: coverage 0.667, more than 28 points below target. Chronos-2 falls short on RMSE (7.071) and coverage (0.814), expected for a zero-shot model on this domain, but produces sharper upper-path estimates than DeepAR ($\mathrm{pinball}_{q95}$ 0.728 vs. 0.817).

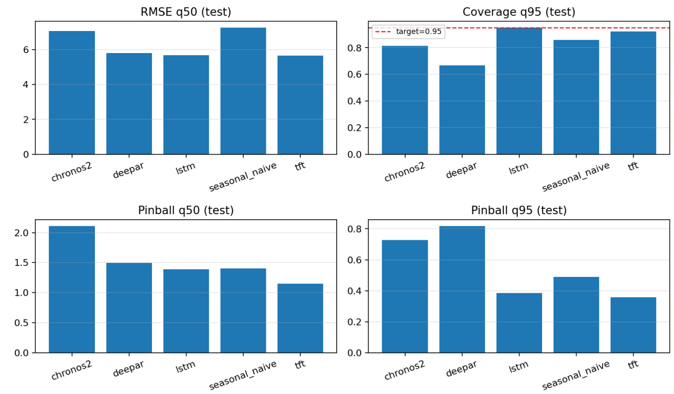


Fig. 2. Forecast metrics on the TIM Milan test set. Left: RMSE and pinball at $q_{0.5}$. Right: pinball at $q_{0.95}$ and coverage. The 0.95 target is shown as a dashed line on the coverage panel.

TABLE IV
SYSTEM METRICS ON THE TIM MILAN TEST SET. REL. SHARP. $= (c_{\mathrm{safe}} - \mathrm{PEAK})/\mathrm{PEAK}$; LOWER IS BETTER. COVERAGE TARGET: $\geq$0.95.

| Model | $\mathrm{MAE}_{\mathrm{CP}}$ | Tol. hit | Coverage | Rel. Sharp. |
|---|---|---|---|---|
| LSTM | **7.175** | **0.487** | **0.984** | 0.237 |
| TFT | 7.386 | 0.462 | 0.968 | 0.153 |
| Seasonal Naive | 7.333 | 0.427 | 0.973 | 0.251 |
| Chronos-2 | 7.415 | 0.420 | 0.873 | **0.082** |
| DeepAR | 7.162 | 0.427 | 0.802 | $-0.041$ |

### B. System quality

Table IV and Figure 3 show the system results. Three findings dominate.

**Coverage vs. relative sharpness tradeoff.** No model dominates both metrics. LSTM maximizes coverage (0.984) with the widest ceiling (relative sharpness 0.237: ceiling is 23.7% above the realized demand peak). Chronos-2 achieves the tightest ceiling in the benchmark (relative sharpness 0.082: only 8.2% above peak) — 65% tighter than LSTM relative to demand — at the cost of coverage (0.873). TFT occupies the best trained operating point: coverage 0.968 (above the 0.95 target), relative sharpness 0.153 (35% tighter than LSTM). Seasonal Naive is safe but wasteful: coverage 0.973, relative sharpness 0.251 (25.1% excess). The tradeoff is real and operators must choose a point on it based on their provisioning risk tolerance. Figure 4 illustrates this tradeoff on individual windows: the Chronos-2 ceiling sits closer to the realized traffic peak with no Milan-specific training.

**CP timing.** All models fall within 0.25 steps of each other ($\mathrm{MAE}_{\mathrm{CP}}$ range 7.16–7.42). Chronos-2 zero-shot is within 0.24 steps of LSTM. Tolerance hit rates range from 0.420 to 0.487: roughly half of all predicted breaks land within 30 minutes of the true break across all models.

**DeepAR failure.** Relative sharpness $-0.041$ means the safe ceiling falls 4.1% below the realized demand peak — the pipeline under-provisions the network. This is not a conservative failure; it is the opposite. The cause is Gaussian confidence intervals that are too narrow for the heavy-tailed traffic

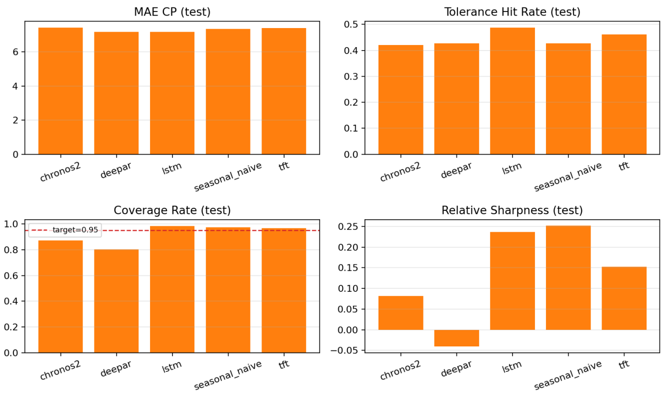


Fig. 3. System metrics after break-point detection and safe-ceiling construction. The coverage panel shows the 0.95 target as a dashed line. DeepAR is the only model with negative sharpness (ceiling below realized traffic peak).

distribution, confirmed by the 0.667 q95 forecast coverage. For safety-critical provisioning, negative relative sharpness is disqualifying.

### C. Post-hoc $\tau$ calibration (negative result)

Post-hoc calibration of $\tau_{\text{pred}}$ was evaluated using random forests, gradient boosting, and per-series offset corrections on validation windows. All calibrators produce zero improvement on test data. After the CLASP validation sweep, the detector is already at near-optimal performance — no residual calibration signal exists. This negative result is informative: it confirms that the sweep-selected CLASP configuration is sufficient and further post-processing does not help.

### D. Pipeline reusability

In operational networks, forecasting models are periodically retrained or replaced as traffic patterns evolve. A pipeline that couples tightly to a single model architecture becomes a liability whenever the model changes. The LLM6G pipeline avoids this by defining a minimal shared interface: any forecaster that produces $(q_{0.5}, q_{0.95})$ over a horizon plugs in. The five-model benchmark demonstrates this property directly — different forecasters produce different points on the coverage vs. relative sharpness tradeoff curve, and this tradeoff is a property of the forecaster and the traffic distribution, not of the pipeline. Operators with strict SLA requirements (coverage $> 0.95$) should prefer LSTM or TFT. Operators optimizing for resource efficiency who can tolerate a coverage margin reduction should prefer Chronos-2.

### E. Chronos-2 and the LLM6G vision

The central result for the LLM6G framing is that Chronos-2 achieves the tightest safe ceiling across all five models without any domain training. Under recurring distribution shifts, operator-grade ML deployment requires retraining cycles whenever the traffic regime changes significantly. Foundation models sidestep this: Chronos-2 enters the control loop immediately with a documented coverage tradeoff (0.873 vs. LSTM's 0.984). Whether that tradeoff is acceptable depends on deployment risk tolerance.

The CP timing gap confirms that Chronos-2 is competitive at regime boundary detection: 7.415 $\text{MAE}_{\text{CP}}$ vs. 7.175 for LSTM, a difference of 0.24 steps (2.4 minutes). Roughly 42% of Chronos-2 break predictions land within 30 minutes of the true break.

### F. TFT and channel-selective normalization

TFT is the best trained model in the benchmark, enabled by a non-trivial implementation decision. Multi-channel input normalization must be applied selectively: raw traffic on channel 0 is normalized by instance mean, but time-feature channels encoding cyclic calendar position must not be rescaled. Normalizing all channels erases the calendar signal — the validation-to-test RMSE gap exceeded 30% under incorrect normalization, and is at 2% with normalization. With channel-selective normalization, TFT achieves RMSE 5.658 and relative sharpness 0.153.

### G. DeepAR as a cautionary case

DeepAR's negative relative sharpness makes it a documented failure case for this pipeline. The model architecture is sound; the failure comes from likelihood mismatch. Milan traffic values are continuous grid-cell aggregates, not integer counts, yet the Gaussian assumption still produces overconfident intervals. This case is instructive for practitioners: choosing a likelihood family is a design decision that propagates through the entire control loop, and the failure is invisible if evaluated only on distributional fit (NLL) rather than on the downstream control metric. The practical lesson: interval calibration must be verified on the control metric (sharpness, coverage), not just on the distributional fit metric.

### H. Context length and stationarity

The 144-step (24-hour) context captures the full daily cycle, which is the dominant periodicity in urban telecom traffic. Shorter contexts miss this cycle. Longer contexts risk including multiple regime changes in the conditioning window, which degrades instance normalization and forces the model to handle multi-regime inputs. The 144-step context is appropriate for this data.

### I. Limitations

The benchmark uses 200 cells, a neutral ascending-ID subset of the full 8,883-cell TIM Milan dataset. All claims are at the level of public grid-cell telecom proxies, not operator base-station telemetry. The evaluation is offline; no real-time receding-horizon controller is implemented.

## VII. Conclusions and Future Works

Proactive control in mobile networks requires forecasting not just traffic levels but the next regime boundary. This paper addressed that need by presenting LLM6G: a shift-aware forecast-to-control pipeline that converts any probabilistic forecast into a safe operating ceiling over the predicted stationary interval and refreshes after the break. Any forecaster

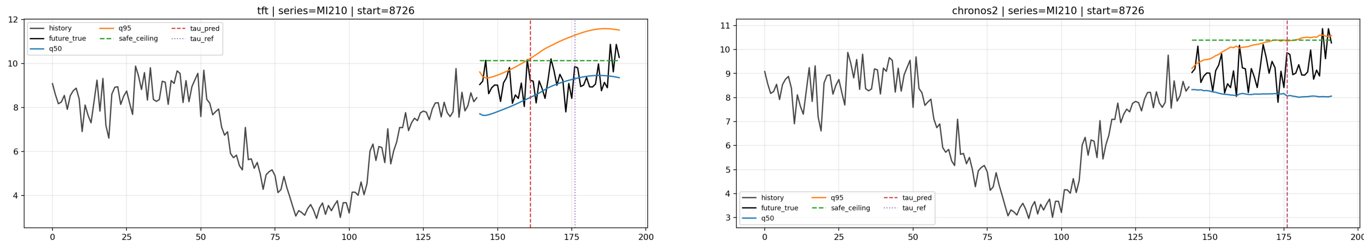


Fig. 4. Example system windows on the test set. Left: TFT (coverage 0.968, rel. sharpness 0.153). Right: Chronos-2 (coverage 0.873, rel. sharpness 0.082). Both show history, forecast quantiles, predicted break, true break, and safe ceiling. The Chronos-2 ceiling sits closer to the realized traffic peak.

that produces $(q_{0.5}, q_{0.95})$ over a horizon plugs into the same interface.

Experimental results on the public TIM Milan dataset with five forecasters confirm that coverage and relative sharpness are competing objectives. TFT achieves the best trained operating point (coverage 0.968, relative sharpness 0.153). Chronos-2, with no domain training, delivers the tightest ceiling in the benchmark, demonstrating that LLM-based foundation models can operate directly in 6G-oriented control loops without retraining. DeepAR documents what happens when likelihood mismatch propagates through the pipeline: the ceiling falls below realized demand.

Two implementation decisions determined whether the loop works or fails silently: channel-selective normalization for TFT, and quantile-based checkpoint monitoring. These decisions are documented as reproducible contributions.

Three directions follow from this work. First, scaling to the full 8,883-cell benchmark tests whether the findings hold at full city scale. Second, evaluation on operator-grade sector telemetry tests the transfer from public proxy to production data. Third, coverage-constrained fine-tuning of Chronos-2 would test whether the foundation model can close the coverage gap while retaining its sharpness advantage — a direct operationalization of the LLM6G vision.

## REFERENCES


[1] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

[2] W. Jiang, "Cellular traffic prediction with machine learning: A survey," *Expert Systems with Applications*, vol. 201, p. 117163, 2022.

[3] E. Lykakis, I. O. Vardiambasis, and E. Kokkinos, "Data traffic prediction for 5g and beyond: Emerging trends, challenges, and future directions: A scoping review," *Electronics*, vol. 14, no. 23, p. 4611, 2025.

[4] N. Rajule, M. Venkatesan, R. Menon, and A. Kulkarni, "Network traffic prediction with reduced power consumption towards green cellular networks," *International Journal of Computer Network and Information Security*, vol. 15, no. 6, pp. 64–77, 2023.

[5] F. Rau, I. Soto, D. Zabala-Blanco, C. Azurdia-Meza, M. Ijaz, S. Ekpo, and S. Gutierrez, "A novel traffic prediction method using machine learning for energy efficiency in service provider networks," *Sensors*, vol. 23, no. 11, p. 4997, 2023.

[6] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley, 2015.

[7] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.

[8] D. Salinas, V. Flunkert, and J. Gasthaus, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *arXiv preprint arXiv:1704.04110*, 2017.

[9] S. Althamary *et al.*, "Enhanced multi-task traffic forecasting in beyond 5g networks: Leveraging transformer technology and multi-source data fusion," *Future Internet*, vol. 16, no. 5, p. 159, 2024.

[10] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. P. Arango, S. Kapoor, J. Zschiegner, D. Maddix, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," *arXiv preprint arXiv:2403.07815*, 2024.

[11] A. F. Ansari *et al.*, "Chronos-2: From univariate to universal forecasting," *arXiv preprint arXiv:2510.15821*, 2025.

[12] R. Koenker and G. Bassett, "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.

[13] P. F. Christoffersen, "Evaluating interval forecasts," *International Economic Review*, vol. 39, no. 4, pp. 841–862, 1998.

[14] Y. Zhu and S. Wang, "Joint traffic prediction and base station sleeping for energy saving in cellular networks," in *IEEE International Conference on Communications*, 2021.

[15] J. Lin, Y. Chen, H. Zheng, M. Ding, P. Cheng, and L. Hanzo, "A data-driven base station sleeping strategy based on traffic prediction," *IEEE Transactions on Network Science and Engineering*, 2024.

[16] M. Masoudi *et al.*, "Digital twin assisted risk-aware sleep mode management using deep q-networks," *arXiv preprint arXiv:2208.14380*, 2022.

[17] G. Barlacchi, M. D. Nadai, R. Larcher, A. Casella, C. Chitic, G. Torrisi, F. Antonelli, A. Vespignani, A. Pentland, and B. Lepri, "A multi-source dataset of urban life in the city of milan and the province of trentino," *Scientific Data*, vol. 2, p. 150055, 2015.

[18] M. Ayaz, D. Fernandez, and J.-C. Juan, "From patterns to predictions: Spatiotemporal mobile traffic forecasting using AutoML, TimeGPT and traditional models," *Computers*, vol. 14, no. 7, p. 268, 2025.

[19] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[20] S. Zhao, X. Jiang, G. Jacobson, R. Jana, W.-L. Hsu, R. Rustamov, M. Talasila, S. A. Aftab, Y. Chen, and C. Borcea, "Cellular network traffic prediction incorporating handover: A graph convolutional approach," in *IEEE International Conference on Sensing, Communication, and Networking (SECON)*, 2020.

[21] J. Wang, L. Shen, and W. Fan, "A TSENet model for predicting cellular network traffic," *Sensors*, vol. 24, no. 6, p. 1713, 2024.

[22] V. Perifanis, N. Pavlidis, S. F. Yilmaz, F. Wilhelmi, E. Guerra, M. Miozzo, P. S. Efraimidis, P. Dini, and R.-A. Koutsiamanis, "Towards energy-aware federated traffic prediction for cellular networks," in *8th International Conference on Fog and Mobile Edge Computing (FMEC)*, 2023, pp. 93–100.

[23] R. Killick, P. Fearnhead, and I. A. Eckley, "Optimal detection of changepoints with a linear computational cost," *Journal of the American Statistical Association*, vol. 107, no. 500, pp. 1590–1598, 2012.

[24] A. Ermshaus, P. Schäfer, and U. Leser, "ClaSP: Parameter-free time series segmentation," in *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, 2022, pp. 3787–3791.